# SaveIQ – Smart Savings Engine

## 1. Overview

SaveIQ is a web-based personal savings tracker. Users define savings goals, log contributions, monitor progress against deadlines, and get AI-generated guidance. The app is backed by **Google Sheets** for data storage and **Groq (LLaMA 3.3-70B)** for the AI Coach. Automated alerts are delivered through **Google Apps Script** time-driven triggers and Gmail.

The header shows a **"Sheets & Groq Sync Active"** status badge on every page, and the sidebar footer carries a **Bank-Grade Encryption / Google Sheets & Groq Sync** note.

## 2. Application Layout

**Sidebar (left)**

- SaveIQ logo and tagline ("Smart Savings Engine")
- **Add New Goal** primary button
- Navigation: Dashboard, My Goals, AI Coach (tagged *LLAMA*), Analytics, Notifications (unread-count badge), Profile, Settings

**Top bar**

- Breadcrumb (`SAVEIQ / <Page>`)
- Sync status badge
- Dark-mode toggle
- Notification bell (with unread indicator)
- **+ New Goal** button
- User avatar and name

The design uses a green accent palette, rounded cards, and a responsive layout (the top bar and cards reflow at narrower widths).

## 3. Dashboard

A greeting ("Good evening, <name>") with an at-a-glance summary of savings progress. Two actions sit beside the heading: **Run Daily Audit** and **Add Goal**.

### 3.1 Summary cards

| Card | Meaning | Demo value |
|---|---|---|
| Total Savings | Accumulated wealth across all goals | ₹190,000 |
| Total Target | Combined target across all goals | ₹380,000 |

| Card | Meaning | Demo value |
|---|---|---|
| Overall Progress | Weighted average completion | 50% |
| Active Goals | Goals in progression | 4 |
| Completed Goals | Goals that reached 100% | 0 |
| Overdue Goals | Goals needing a pace adjustment | 0 |

### 3.2 Portfolio Completion Rate

A live-aggregated progress bar (₹190,000 / ₹380,000, 50% completed, ₹190,000 remaining). Beside it, a **SaveIQ Coach** card gives a one-line pace summary (e.g., "on track for 3 of 4 milestones") with an **Ask AI Coach** link into the AI Coach page.

### 3.3 Active Goals and Recent Activity

- **Active Goals (4)** shows compact goal cards (category, name, saved/target, progress bar, due date) with a quick **+ Add Savings** button and a **View All** link to My Goals.
- **Recent Activity** lists the latest contributions with date, note, and amount (for example, Emergency Fund +₹15,000, Travel Fund +₹8,000, Education Fund +₹12,000, Emergency Fund +₹10,000).

## 4. My Goals

Title: **My Savings Goals** – *track individual goal velocity, monitor deadlines, and add savings contributions.*

### 4.1 Controls

- **Search** box
- **Status filter tabs:** All, Active, Completed, Overdue
- **Category dropdown** (All Categories)
- **Sort dropdown** (e.g., Earliest Deadline)
- **Create New Goal** button

### 4.2 Goal cards

Each card shows category, goal name, short description, status tag (Active), **Saved** vs **Target**, a progress bar, and three metrics: **Remaining**, **Days Left**, **Monthly Target**. Card actions: **view** (eye), **edit** (pencil), **delete** (bin), and **+ Add Savings**.

### 4.3 Demo goals

| Goal | Category | Saved | Target | Progress | Remaining | Days left | Monthly target |
|---|---|---|---|---|---|---|---|
| Education Fund | Education | ₹48,000 | ₹80,000 | 60% | ₹32,000 | 30 | ₹16,000 |

| Goal | Category | Saved | Target | Progress | Remaining | Days left | Monthly target |
|---|---|---|---|---|---|---|---|
| Travel Fund | Travel | ₹32,000 | ₹50,000 | 64% | ₹18,000 | 46 | ₹9,000 |
| Emergency Fund | Emergency | ₹65,000 | ₹100,000 | 65% | ₹35,000 | 92 | ₹10,000 |
| Vehicle Fund | Vehicle | ₹45,000 | ₹150,000 | 30% | ₹105,000 | 197 | ₹15,000 |

## 5. AI Coach

**SaveIQ Coach** provides personalized mathematical analysis and behavioral guidance for the user's goals. It is powered by **Groq LLaMA 3.3-70B**, and a counter shows how many goals are in the evaluation context (4 in the demo).

### 5.1 Select Analysis Objective (quick actions)

1. **Analyze My Progress**
2. **Create a Savings Plan**
3. **How Can I Reach My Goal Faster?**
4. **Explain My Financial Progress**
5. **Give Me Saving Suggestions**

### 5.2 Custom questions

A free-text field accepts specific questions (placeholder example: *"Can I afford ₹15k monthly for Travel without delaying Emergency…"*). The **Analyze** button is disabled until a question is typed. The field highlights with a green outline on focus.

### 5.3 Empty state

Before a request, the panel reads **"Ready to analyze your savings strategy"** with a prompt to click a quick action or ask a question, plus an **Analyze My Progress Now** button. Results are described as real-time financial trajectory feedback.

## 6. Analytics

Title: **Financial Analytics & Projections** – *visual breakdown of savings growth, goal velocity, and category allocations.* A time-range selector offers **3M, 6M, 1Y, ALL**.

### 6.1 KPI cards

| KPI | Demo value | Note |
|---|---|---|
| Total Saved | ₹190,000 | 50% of portfolio target |
| Total Commitment | ₹380,000 | Across 4 target milestones |
| Active Goals | 4 active | 0 completed · 0 overdue |

| KPI | Demo value | Note |
|---|---|---|
| Monthly Contribution | ₹22,500/mo | Historic deposit average |

### 6.2 Charts

- **Goal Progress Comparison** – horizontal bars of current savings vs target for each goal (with percentages).
- **Category Distribution** – donut chart of allocation by category. Demo split: Emergency 34% (₹65,000), Education 25% (₹48,000), Vehicle 24% (₹45,000), Travel 17% (₹32,000).
- **Monthly Savings Velocity** – total deposited per calendar month (month axis, e.g., Aug, Sep). The chart area appeared empty in the recording.
- **Portfolio Status Breakdown** – Completed, Active & Pacing, and Overdue Deadlines counts with percentages (demo: 0 / 4 (100%) / 0).

## 7. Notifications & Audit Engine

Automated alerts are generated by **Google Apps Script time-driven rules** and sent via **Gmail services**.

### 7.1 Actions

- **Preview Gmail Emails** – preview the emails the engine would send.
- **Run Daily Audit Now** – trigger the audit manually.
- **Mark all as read**

### 7.2 Filter tabs

All, Unread, Deadlines, Milestones (counts shown, e.g., All (3), Unread (2)).

### 7.3 Alert types in the demo

| Type | Goal | Example message |
|---|---|---|
| Upcoming Deadline | Education Fund | Deadline approaching on Oct 30, 2026; ₹32,000 remaining |
| Monthly Progress Milestone | Emergency Fund | 65% of the goal reached |
| Travel Goal Deadline | Travel Fund | On track for November departure; ₹18,000 to reach target |

Each alert has a timestamp and a mark-as-read tick.

## 8. Profile (User Financial Profile)

Title: **User Financial Profile** – *customize currency preferences, income benchmarks, and alert schedules.*

| Field | Description | Demo value |
|---|---|---|

| Field | Description | Demo value |
|---|---|---|
| Full Name* | Display name | (user's name) |
| Email Address* | Account email, used for alerts | (user's email) |
| Base Currency* | Dropdown | Indian Rupee (₹) |
| Monthly Income (₹) | Income benchmark | 85,000 |
| Overall Monthly Saving Target (₹) | Portfolio-wide monthly goal | 25,000 |
| Reminder Schedule | Audit frequency dropdown | Daily 8:00 AM Audit (Recommended) |
| Notification Level | Radio cards | All Alerts / Important Only / Muted |

**Notification levels**

- **All Alerts** – deadlines, pace warnings, and milestone celebrations
- **Important Only** – only overdue goals and completion milestones
- **Muted** – no automated reminders or email dispatches

Changes are committed with the **Save Changes** button. The avatar card shows the member tier ("SaveIQ Standard Member").

## 9. Typical User Workflows

1. **Create a goal** – click *Add New Goal* (sidebar), *+ New Goal* (top bar), or *Create New Goal* (My Goals).
2. **Log a contribution** – click *+ Add Savings* on a goal card (Dashboard or My Goals).
3. **Check overall health** – open the Dashboard and review summary cards and Portfolio Completion Rate.
4. **Get advice** – open AI Coach, pick a quick action or type a question, and press Analyze.
5. **Review trends** – open Analytics and change the range (3M / 6M / 1Y / ALL).
6. **Tune alerts** – set Notification Level and Reminder Schedule in Profile, then Save Changes.
7. **Run an audit manually** – use *Run Daily Audit* (Dashboard) or *Run Daily Audit Now* (Notifications).

## 10. Technology Summary

| Component | Role |
|---|---|
| Google Sheets | Data store for goals, contributions, and profile ("Sheets Sync") |
| Groq LLaMA 3.3-70B | AI Coach analysis and suggestions |
| Google Apps Script | Time-driven audit and alert rules |

| Component | Role |
|---|---|
| Gmail | Email delivery of alerts |

## 11. Observations and Gaps

- **Settings** appears in the navigation but was not opened in the recording, so it is undocumented here.
- The **Add Goal / Add Savings forms** and the **AI Coach results** were not shown on screen.
- The Analytics **Monthly Contribution (₹22,500/mo)** differs from the Profile **monthly saving target (₹25,000)**; the first is a historic average and the second is a target.
- The **Monthly Savings Velocity** chart appeared blank in the recording; worth verifying that it populates from Sheets data.
- The recording's audio could not be transcribed, so this document reflects on-screen content only.